

BUSINESS CASE

Digital Marketing : Campaign ROI & Budget Allocation

1. Executive Summary

This analysis answers all 6 questions from the original brief: overall ROMI, ROMI by campaign, performance by date, weekday/weekend revenue, best-performing category, and geo targeting. Overall ROMI is 1.40x. YouTube (3.77x) and the Influencer category (2.54x) are the strongest performers; Facebook (0.66x) and the Social category (0.86x) are the weakest and Social is also the single largest spend category, meaning the most budget currently goes to the worst-performing type. Tier1 geo targeting (1.35x) meaningfully outperforms tier2 (0.72x).

Recommendation: reallocate budget from Social/Facebook toward YouTube and Influencer channels, and prioritize tier1 geo targeting over tier2.

2. Business Problem

Marketing budget is being allocated across campaigns, categories, and geo tiers without a clear, verified view of which ones actually generate a return.

3. Data & Method

309 daily campaign records were used, covering spend, revenue, clicks, leads, and orders by campaign, category, date, and geo tier. One row was excluded from every calculation below see Section 4 for what it was and why it doesn't affect the reliability of these results.

Tool choice: this project uses SQL and Excel only, not Python. Every question here is a direct aggregation (totals, averages, ratios by group) none require formal statistical testing (e.g. ANOVA, chi-square, correlation), which is where Python is reserved elsewhere in this portfolio (Car Insurance, Customer Churn, Hotel Bookings, Amazon Sales). SQL retrieves and aggregates the data; Excel builds on the same numbers for the pivot analysis, the what-if budget model, and the dashboard so every figure in this document can be checked two independent ways without a third tool duplicating work neither one already does well.

4. A Data Issue Found in the Source File (Not This Analysis)

One row in the raw data (mark spent = ₦0, revenue = ₦42,889,366) does not represent a real campaign. Tracing it back to the original source spreadsheet confirms exactly what it is: a spreadsheet TOTAL row (labelled "Total" in the id column), where someone had summed the whole campaign list for their own reference. Its spend and revenue figures match the sum of all 308 real rows exactly.

This is a source-file issue, not something introduced during this analysis, and not something a downstream analyst can correct at the root the fix belongs upstream, in how the spreadsheet or data export is built (a summary row should never be exported alongside transaction-level records). The correct and complete downstream response, applied throughout this project, is to identify it and exclude it from every calculation, which is what was done here. Left uncorrected, it would have inflated the headline ROMI from the true 1.40x to a misleading 2.80x.

5. Key Findings :All 6 Brief Questions

Q1. Overall ROMI

1.40x (₦42.89M revenue on ₦30.59M spend), after excluding the Total row described above.

Q2. ROMI by campaign

Campaign	Spend	ROMI
YouTube	₦4.06M	3.77x
Instagram	₦7.88M	1.40x
Banner	₦5.03M	1.22x
Google	₦3.46M	1.07x
Facebook	₦10.17M	0.66x

Q3. Performance by date

Both the highest-spend day and the highest-revenue day are 2021-02-20 (₦3.50M spend, ₦5.26M revenue). Average order value stays consistent across the top 5 spend days (₦4,692–₦4,760). The highest-conversion day was 2021-02-10 (2.83% leads-per-click); the lowest was 2021-02-02 (1.29%) more than double the gap between best and worst day.

Q4. Weekday vs. weekend average revenue

Weekday: ₦141,914 average daily revenue. Weekend: ₦132,594. A real but modest gap (about 7% higher on weekdays)not a basis for a major schedule change on its own.

Q5. Which category works best

Category	Spend	ROMI
Influencer	₦8.31M	2.54x
Media	₦5.03M	1.22x
Search	₦3.46M	1.07x
Social	₦13.80M	0.86x

Social is both the worst-performing category and the largest spend category the most money is going to the weakest type.

Q6. Geo targeting: tier1 vs. tier2

Tier1: 1.35x ROMI (₦5.13M spend). Tier2: 0.72x ROMI (₦5.76M spend). Checked month-by-month as well as in aggregate the gap holds consistently, not just as a one-time snapshot.

6. Recommendation

Reallocate budget from Facebook and the Social category toward YouTube and Influencer campaigns, which are already proving 2–6x more efficient with comparable or less spend.

Prioritize tier1 geo targeting over tier2 in future campaigns the gap is consistent, not a fluke.

Add a standing data-quality check to the reporting pipeline (flagging any row where spend = 0 but revenue is implausibly large) so a future refresh of this dataset can't reintroduce the Total-row error silently this is a downstream safeguard, since the actual fix belongs with whoever maintains the source spreadsheet.

7. Glossary of Abbreviations

Full meanings of the abbreviations used throughout this document and the underlying analysis:

Abbreviation	Full Meaning
ROMI	Return On Marketing Investment : how much revenue was generated per unit of money spent
CAC	Customer Acquisition Cost: the average marketing spend needed to acquire one paying customer
CPC	Cost Per Click :the average amount spent for each click an ad receives
CPL	Cost Per Lead : the average amount spent to generate one lead (a person who signed up but hasn't yet purchased)
AOV	Average Order Value : the average amount of revenue generated per completed order
CTR	Click-Through Rate : the percentage of people who saw an ad and actually clicked on it
Conversion 1	The rate at which website visitors turn into leads ($\text{Leads} \div \text{Clicks}$)
Conversion 2	The rate at which leads turn into paying customers ($\text{Orders} \div \text{Leads}$)